LIM-2.D.1 The concept of a limit can be extended to include infinite limits.

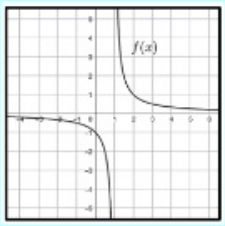
LIM-2.D.2 Asymptotic and unbounded behavior of functions can be described and explained using limits.

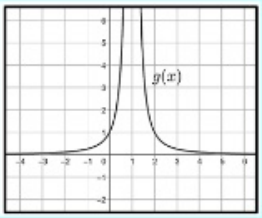
1-14 Infinite Limits and Vertical Asymptotes

Get Ready...

Use the graphs of $f(x)$ and $g(x)$ below to find the following limit values if possible.

$f(x) = \dfrac{1}{x-1}$ $\qquad$ $g(x) = \dfrac{1}{(x-1)^2}$





A. $\lim\limits_{x \to 1^-} f(x) =$ $-\infty$

B. $\lim\limits_{x \to 1^+} f(x) =$ $+\infty$

C. $\lim\limits_{x \to 1} f(x) =$ DNE

D. $\lim\limits_{x \to 1^-} g(x) =$ $+\infty$

E. $\lim\limits_{x \to 1^+} g(x) =$ $+\infty$

F. $\lim\limits_{x \to 1} g(x) =$ $+\infty$

THINK ABOUT IT: What do limits that result in infinity or negative infinity tell us about a function?

Important Facts to remember			
$\frac{\#}{0} \to +\infty$ or $-\infty$	$\frac{\#}{\infty} \to 0$	$\frac{\infty}{\#} \to +\infty$ or $-\infty$	$\frac{0}{0}$ and $\frac{\infty}{\infty}$ are indeterminate forms

Definition of Vertical Asymptote	If $\lim_{x \to c^-} f(x) = \pm\infty$, $\lim_{x \to c^+} f(x) = \pm\infty$, or $\lim_{x \to c} f(x) = \pm\infty$ then $f(x)$ has a vertical asymptote at $x = c$.

RECALL: A vertical asymptote will ONLY occur when a factor in the denominator does NOT cancel out!

Example 1: Determine all vertical asymptotes for the functions below.

a. $f(x) = \dfrac{x^2+1}{x^2-1}$

$= \dfrac{x^2+1}{(x+1)(x-1)}$

VA: $x+1=0$ $\quad x-1=0$

$x=-1$ $\quad x=1$

b. $f(x) = \dfrac{x^2+2x-8}{x^2-4}$

$= \dfrac{(x+4)\cancel{(x-2)}}{(x+2)\cancel{(x-2)}}$

VA: $x+2=0$

$x=-2$

Find the following limits, if possible.

Example 2: $\lim_{x \to 1} \frac{2}{x-1}$ = DNE

$-\infty \quad +\infty$

$.9 \quad 1 \quad 1.1$

Find the following limits, if possible.

Example 3: $\lim_{x \to 3} \frac{x^2-4x+3}{x^2+2x-15}$

$\frac{9-12+3}{9+6-15} = \frac{0}{0}$

$= \lim_{x \to 3} \frac{\cancel{(x-3)}(x-1)}{(x+5)\cancel{(x-3)}}$

$= \lim_{x \to 3} \frac{x-1}{x+5}$

$= \frac{2}{8} = \frac{1}{4}$

Find the following limits, if possible.

Example 4: $\lim_{x \to 2^+} \frac{x-5}{x-2} = -\infty$

Find the following limits, if possible.

Example 5: $\lim_{x \to -3^-} \frac{3x+10}{x+3} = -\infty$

$\frac{1}{0}$

$-\infty$

-3.1 -3

$\frac{+}{-}$

Example 6: Determine which function(s) below have a vertical asymptote based on the limit statements.

a. $\lim_{x \to 3} f(x) = 0$ no

b. $\lim_{x \to 2} g(x) = \infty$ yes $x=2$

c. $\lim_{x \to -5^+} h(x) = \infty$ yes

$x = -5$

d. $\lim_{x \to \infty} k(x) = -4$ no

Example 7: Consider the function $k(x) = \frac{x^2+x-20}{x^2-b}$ where b is a nonzero constant.

a. Find a possible value of b such that the function $k(x)$ has exactly one vertical asymptote.

$$K(x) = \frac{(x+5)(x-4)}{(x+\sqrt{b})(x-\sqrt{b})}$$

$x+\sqrt{b} = x+5$ OR $x-\sqrt{b} = x-4$

$\sqrt{b} = 5$ $\quad$ $-\sqrt{b} = -4$

$b = 25$ $\quad$ $\sqrt{b} = 4$

$b = 16$

Example 7: Consider the function $k(x) = \frac{x^2+x-20}{x^2-b}$ where b is a nonzero constant.

b. Find a possible value of b such that the function $k(x)$ has two vertical asymptotes.

$k(x) = \frac{(x+5)(x-4)}{(x+\sqrt{b})(x-\sqrt{b})}$

Any number except 16 or 25

Example 7: Consider the function $k(x) = \frac{x^2+x-20}{x^2-b}$ where b is a nonzero constant.

c. Find a possible value of b such that the function $k(x)$ has no vertical asymptotes.

$$k(x) = \frac{(x+5)(x-4)}{(x+\sqrt{b})(x-\sqrt{b})}$$

$$\boxed{b<0}$$

$x^2-b \neq 0$

$b<0$ $b=-3$

x^2+3